

Supplementary Methods

S1. Exact Computational Environment and Model Provenance

Table S1 records the executable environment used or recovered for the analyses. The vector and result artifacts are hash-addressed. The BGE-M3 vectors used in the article come from a single recorded run over corpus v2 (contract chinese-rap-repaired-corpus-embeddings-v2/1.0.0): Tesla T4, half precision, batch size 8, maximum length 2,048, corpus source order, checkpoint weights SHA-256 b5e0ce34...6aad38, corpus content SHA-256 8102da32...4b313b; a resume under any other device, precision, weights, or corpus is refused. The earlier corpus's vectors had been produced under an unrecorded device and precision; the configuration probe (results/embedding-configuration-probe-v1) found that three candidate configurations all reproduce them within a cosine tolerance of 2.2×10^{-5} , which bounds the concern without identifying the configuration, and the single recorded run replaces them.

Component	Exact record	Analysis-relevant settings	Evidence status
Execution environment	CPython 3.12.13; Windows 11 (10.0.26200)	Task-specific deterministic seeds; Python executable SHA-256 e062889e...aea35	Direct for BGE, named-entity recognition (NER), and rhyme; retrieval manifest lacks a contemporaneous runtime lock
Semantic encoder	BAAI/bge-m3 revision 5617a9f61b028005a4858fdac845db406aefb181; FlagEmbedding 1.4.0; PyTorch 2.11.0+cu128; Tesla T4	Dense head; 1,024 dimensions; max_length=2048; batch 8; use_fp16=True; L2 normalization	Single recorded run over corpus v2; checkpoint weights, corpus digest, device, and precision pinned in the run contract; vector file SHA-256 247e9619...51da1
NER model	ckiplab/albert-tiny-chinese-ner revision bcb519856ca93a666b1e48a9daef3f88c9b572a0; Transformers 5.14.1; PyTorch 2.11.0+cu128	180-character windows; 40-character overlap; maximum 256 tokens; batch 96	Runtime, checkpoint, weights, and vocabulary hashes verified

Component	Exact record	Analysis-relevant settings	Evidence status
Written-ending extraction/model	pypinyin 0.55.0; scikit-learn 1.9.0	<code>Style.FINALS_TONE3; strict=True; neutral tone 5; averaged log-loss SGD</code>	Direct package-version and module-hash evidence
Numerical/data stack	NumPy 2.4.6; pandas 3.0.5; SciPy 1.18.0; scikit-learn 1.9.0	Sparse term frequency–inverse document frequency (TF–IDF), normalization, tests, tables	Direct for NER/rhyme; forensic rather than manifest-bound for retrieval
Transformer support stack	Transformers 5.14.1; tokenizers 0.22.2; huggingface-hub 1.27.0	Local/offline model loading	Transformers direct for NER; remaining BGE support versions recovered locally

The BGE checkpoint configuration contains `transformers_version=4.33.0`; that is upstream serialization metadata and is not reported as this project's runtime. The canonical environment-map fingerprint is SHA-256 `b290363fa776f50219cd16ee124fd88b7253d77658bdb9f0d2cef42ee2453152`. Full file hashes and official artifact URLs appear in `software_citation_recommendations.md` in the release audit directory.

S2. PD-001 Retrieval Protocol Amendment

The original frozen research contract required vectorizers to be fitted on training data only. The released character TF–IDF builder instead calls `fit_transform` once on all document text in the fixed retrieval corpus before constructing group-held-out label profiles. Thus each query affects the unlabeled character vocabulary and inverse-document-frequency distribution, although its source label, relevance status, rank, and evaluation outcome are not supplied to the vectorizer. The equal fusion weight is also not outcome-tuned.

The reported retrieval estimand is therefore:

Within the frozen corpus, when a song's source-credit membership and its contribution to candidate profiles are held out, how highly does the remaining source-label repertoire rank?

It is not a prospective estimate for later songs or another corpus. The original contract remains unchanged; `PROTOCOL_AMENDMENT_PD001_TRANSDUCTIVE_TFIDF.md` records

the audit chronology and affected artifacts. The primary fixed-corpus estimates remain those of the frozen implementation.

A grouped sixfold sensitivity tests whether transductive vocabulary and IDF exposure explains the fusion result. Each of 5,430 global exact/near-duplicate components is assigned wholly to one fold; the 5,455 queries are distributed as 910 in one fold and 909 in each of five folds. Every test fold contains all 204 source-credit labels, and each corresponding training partition retains at least five components per label. Inductive TF-IDF is fitted only on the five training folds and then transforms their held-out fold. A matched transductive system fits vocabulary and IDF on all 5,455 unlabeled texts but uses the same fold-specific training profiles, test queries, duplicate weights, candidate labels, fusion rule, and rankings. Because cross-fitted profiles contain five-sixths of the components, their absolute scores are a sensitivity result rather than a direct replacement for leave-one-component-out Table 2.

Inductive fusion yields macro MRR 0.439712 versus 0.404763 for inductive TF-IDF, a paired difference of +0.034949 (95% CI 0.024373–0.046082). Matched transductive exposure increases TF-IDF MRR by +0.003963 (0.001020–0.006942) and fusion MRR by +0.003681 (0.001413–0.006416). Intervals use the same 5,000-replicate paired two-stage label/component bootstrap principle as the primary analysis. Thus evaluation-corpus exposure is mildly optimistic within this corpus, but its magnitude cannot explain the training-only fusion advantage. This sensitivity does not estimate prospective or external-population performance.

S3. PD-002 Corpus Reconciliation and Upstream Chunk Deduplication

PD-002 was recorded after the frozen analyses when the legacy cleaner was found to delete repeated (source-credit label, exact cleaned chunk text) rows before downstream duplicate controls. The reconciliation exactly reproduces the four historical stages:

26,833 chunks/7,721 song IDs in the raw export; 25,279/7,420 after title exclusions; 25,026/7,391 after line cleaning and empty removal; and 22,132/7,214 after keep-first chunk deduplication. The final stage removes 2,894 chunks and erases 177 song records. Of those records, 131 meet the conservative high-confidence rule of matching source-credit label, normalized title, and complete cleaned chunk sequence; the remaining 46 require review rather than automatic deletion.

The live native Google Sheet and local raw export align on all 26,833 song/chunk keys. All 33/33 substantive cell differences were adjudicated (29 title type coercions, two leading-apostrophe semantics, and two control-character import side effects), with none unresolved; this is a row-aligned provenance check, not a claim of remote/local byte identity. In the released shared-text-exclusion rhyme frame, the 204 labels remain fixed while restoring eligible pre-snapshot chunks expands songs from 5,619 to 5,621, line-bearing songs from 5,452 to 5,459, and transition-bearing songs from 5,347 to 5,353. Without retraining, this adds 7,033 strict-Han-ending lines, 4,938 adjacent transitions, and 6,472 repeat occurrences beyond first occurrences. Aggregate ending-family shares remain stable (total-variation distance 0.001789; switch rate lower by 0.001390, counterfactual minus released), but the duplicate-aware population has not been used to retrain or reevaluate the predictive models. No stability claim is therefore made for top- k accuracy, mean reciprocal rank, calibration, abstention, or paired model differences.

S4. Repertoire-Graph Null and Projection Diagnostics (companion interface only)

The interactive companion's repertoire graph is not used as evidence in the article; this section is retained so that the interface's own diagnostics remain documented. The published graph is the intersection of a 140-edge reciprocal-top-five network under primary duplicate control and a 145-edge network after cross-label shared-text exclusion. The observed intersection contains 86

edges. The primary null holds the first network fixed and applies 1,450 successful undirected double-edge swaps to the second network per replicate, preserving its complete node-degree sequence while randomizing endpoints. Across 10,000 fixed-seed replicates, the null intersection has mean 4.524, median 4, 95% range 1–9, and maximum 13; no replicate reaches 86, giving an add-one Monte Carlo p of 0.00010. An auxiliary 100,000-replicate source-label permutation gives mean 0.979 and $p = 0.000010$. These tests concern cross-treatment adjacency alignment, not edgewise significance, social relations, influence, or external cultural structure.

The deterministic two-dimensional display combines the normalized primary and sensitivity centroids, recentres them, and applies singular-value-decomposition PCA. Its first two components explain 26.20% of variation. Using cosine distance in the original 1,024-dimensional consensus space and Euclidean distance in the display, trustworthiness is 0.785, 0.795, and 0.804 at $k = 5, 10$, and 15. Exact-neighbour retention is 0.173, 0.268, and 0.347, compared with random expectations of 0.025, 0.049, and 0.074. Pairwise distance-rank correlation is 0.680. Only 18 of 86 released edges are mutual top-five neighbours in two dimensions. Accordingly, the PCA is a navigation overview; released graph lines, not screen distance, carry the relation claim.

S5. External Written-Ending Implementation Check

The external MuChin resource (Wang *et al.* 2024; DOI 10.24963/ijcai.2024/860) was used only as an implementation sensitivity check. Across 1,000 publisher-recommended folders, 44,980 written lines align exactly by line and section. Treating exported R markers as same-family-within-section indicators gives an F1 score (the harmonic mean of precision and recall) of 0.931 and a Matthews correlation of 0.541 for the project's deterministic same-family detection.

This is not independent gold. The MuChin interface automatically groups and colour-highlights pinyin rhyme families, annotators principally review polyphonic readings, and the exported $\mathbb{R}$ field collapses family identity. The check is therefore partially circular, excluded from model selection, and does not validate Chinese-rap rhyme, pronunciation, tone, cadence, beat alignment, or flow.

S6. NER Annotation Path to Human Gold

The corpuswide private review package contains 800 distinct occurrences sampled by proposed type and agreement state: exact agreement, overlap agreement, type/boundary conflict, lexicon only, and transformer only. A separate claim-directed package contains every unique occurrence contributing to the ten currently released claims: 102 label/reference occurrence rows and 60 co-mention occurrence rows, with five rows overlapping, for 157 unique occurrences in total. Coverage is therefore 157/157 (100%) within the released-claim frame. Reviewer occurrence sheets add 20 real comparators and 16 synthetic boundary-attention controls; pair sheets add eight real non-released comparators to the 23 released co-mention support tasks.

For both packages, two reviewers work independently; in the claim-directed package, they are blind to claim/control status. Uncertainty, corrections, conflicting link decisions, and all disagreements require adjudication. At release, both reviewers and the adjudication sheets contain zero completed decisions. The claim-directed audit can ultimately confirm or reject the evidence underlying the released edges, but it is claim-conditioned and has no corpuswide negative frame; it cannot estimate corpuswide precision, recall, or F1. The 800-item package remains the route to those metrics and a song-level train/development/test split. Inter-annotator agreement, claim-confirmation rates, and all accuracy metrics remain withheld.

Two earlier, separate sets of 300 context forms and 300 occurrence tasks also contain no completed decisions and are superseded. No unreviewed package supports network centrality or demographic comparisons. The present 22-surface inventory and its six label-to-place and four co-mention edges are reproducibility-screened provisional evidence only.

S7. Public/Private Release Boundary

The public supplement contains aggregate metrics, uncertainty intervals, typed entity surfaces, support summaries, written-ending classes, figure-source tables, code, contracts, and validation manifests. It excludes lyric text, full written lines, song/chunk identifiers, private membership rows, row-level lyric hashes, embeddings, and reviewer contexts. Short Chinese surfaces are included only when needed to interpret an aggregate entity or written-ending result.

The private frozen snapshot is required to rebuild row-level artifacts. Its original platforms, acquisition route and dates, temporal coverage, sampling frame, transcription origin, rights basis, and ethics determination are not established in the supplied export.

`DATA_PROVENANCE_AND_AUTHOR_ACTIONS.md` separates verified local facts from the owner-supplied information required before journal submission.

S8. Leakage Groups, Label Identity Audit, and the Rhyme-Form Space

Leakage groups. Over the 7,236 query songs, two songs share a leakage group when they share a corpus v2 exact-text component or when their whole-song character-trigram sets have Jaccard similarity of at least 0.80 after NFKC normalisation, case folding, and removal of punctuation and spaces; groups are connected components of the union. Neither rule contains the other: 404 of 799 multi-song text components would be split by the near-duplicate rule alone, and 44 near-duplicate pairs share no text component. The union yields 5,888 groups, 798 with more than one song, the largest with 27; every label retains at least five groups (median 33). Each half of the

union is scored alone as an ablation in `results/retrieval-v2/analysis_summary.json`. The module `src/leakage_groups_v2.py` carries fifteen tests, each rule tested alone before the union and the invariant shown to be falsifiable. The written-rhyme partitions use the same unit: assigning songs individually, as the earlier build did, would have placed 433 of 799 multi-song components across a partition boundary, touching 1,330 song records.

Label identity. Among 241 labels, no pair is identical after NFKC normalisation, case folding, and removal of spaces and punctuation, and all twelve substring candidates are stage-name particles (ICE in Ice Paper). Shared passages of at least thirty characters connect 511 label pairs; the highest mutual involvement share is 48.5%, three pairs exceed 30%, none exceeds 50%. The 45 pairs at or above 10% were reviewed by the author and determined to be collaborations. No normalisation or merging is applied. `results/label-identity-v1/` records the measurement and the determination separately.

Rhyme-form space. A song's rhyme-form document is the sequence of tokens emitted for each classifiable line ending: `FAM_x` (one of the seventeen families), `TONE_t`, `PAIR_a>b` (penultimate and last families), `TR_a>b` (previous line's family to this one), `SAME` or `SHIFT`, and one `RUN_k` token per maximal run of same-family endings (buckets 1, 2, 3, 4+). The second tier adds `FIN_`, `FINT_`, and `PFIN_` tokens for the raw final, toned final, and final pair. Tokens are weighted by the lexical system's TF-IDF settings (document frequency at least three, sublinear term frequency, L2 rows), giving 605 and 1,997 features. Songs with fewer than four classifiable endings receive the worst rank in any system that uses this space; 7,129 of 7,236 queries are covered.

Neutralisation arms. Each arm's surfaces are removed from every song before the character space is refitted: Latin surfaces as whole words (a Han character counts as a boundary), Han surfaces as substrings. The reference-core arm has 90 surfaces; places 212 (types GPE, LOC, FAC); named entities 459 (every type except hip-hop slang, English words, and language names); the whole catalogue 605. The dense system is not neutralised because that would require re-embedding, which is a separate GPU run.

S9. The Whitening Probe

Transform. For a training set of song vectors x_i with labels c_i and weights w_i (each (leakage group, label) component weighted one), the within-author covariance is $\Sigma_w = \Sigma_i w_i (x_i - \mu_{c_i})(x_i - \mu_{c_i})^T / \Sigma_i w_i$, with μ_{c_i} the weighted label mean; Ledoit–Wolf shrinkage towards a scaled identity is estimated on the weighted residuals (shrinkage 0.030–0.031 across folds). The transform maps a song vector to $\Sigma_w^{-1/2}(x - \mu)$ by the symmetric (ZCA) square root and re-normalises it; label profiles and cosine ranking then run unchanged. *Total whitening* uses the total covariance about the weighted mean (shrinkage 0.024); *permuted-label whitening* shuffles the training labels before estimating Σ_w ; *centring* subtracts the mean only. Within-author variation is 89.5–89.7% of the total variance across folds, against 95.9–96.1% under the permutation null.

Folds. Leakage groups are dealt into five folds with seed 20260825 (1,408 / 1,492 / 1,507 / 1,382 / 1,447 queries). Every fold's transform is fitted on the other four folds' songs and applied to every song; only the fold's own queries are scored, against profiles built from all other songs exactly as in the main protocol. The dense scorer reproduces the builder's scores to 7×10^{-7} and the untransformed space reproduces the builder's MRR (0.3181) before any transformed space is trusted.

Matched-dimension lexical control. The TF-IDF rows are reduced by a truncated singular-value decomposition to 1,024 components fitted fold-wise on the training songs (43.7–43.8% of variance kept), unit-normalised, and given the same five transforms. Untransformed, the reduced space scores 0.393 (−0.057 against the full 150,000-feature rows, interval [−0.062, −0.052]); total whitening lifts it to 0.511 and within-author whitening to 0.518, +0.008 ([+0.005, +0.011]) over total and +0.009 ([+0.006, +0.012]) over the permutation null. The whitened lexical space exceeds the whitened semantic space by 0.070 ([+0.060, +0.081]). Fusing the two whitened spaces by the same equal-weight z-score rule gives 0.587 (Recall@1 0.482, Recall@10 0.783), +0.042 ([+0.037, +0.047]) over the fusion of raw lexical with whitened semantic and +0.069 ([+0.062, +0.075]) over the whitened lexical space alone.

Transfer. A deterministic 15% of labels (34 of 226, seed 20260825) is held out whole. Transforms are fitted on the other labels' songs only and the held-out labels' queries are scored against the usual leave-group-out profiles of all 226 labels. On those queries the untransformed space scores 0.297; total whitening 0.455 (+0.156, [+0.131, +0.180]); within-author whitening fitted on other authors 0.429 (+0.129 over untransformed, but −0.027 against total, [−0.035, −0.019]); the fold-wise within-author transform that had seen those authors 0.421 (−0.008 against the unseen-author transform, [−0.016, −0.000]).

Averaging. `results/retrieval-v2/identity_estimands.json` reports every system of the identity-space and probe experiments under plain (per-query), component-weighted, and macro (per-label) averaging. Under the macro estimand the raw semantic space scores 0.301, the raw lexical 0.398, within-author whitening 0.440, the whitened lexical space 0.487, the fusion of raw lexical with whitened semantic 0.506, and the fusion of the two whitened spaces 0.564; the ordering of spaces is the same under every average.

Software and Model Records

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<https://huggingface.co/BAAI/bge-m3/tree/5617a9f61b028005a4858fdac845db406aefb181>
- CKIP Lab (2022) *ckiplab/albert-tiny-chinese-ner*, revision bcb519856ca93a666b1e48a9daef3f88c9b572a0 [pretrained token-classification model]. <https://huggingface.co/ckiplab/albert-tiny-chinese-ner/tree/bcb519856ca93a666b1e48a9daef3f88c9b572a0>
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- mozillazg and contributors (2025) *pypinyin*, version 0.55.0 [computer software].
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- The scikit-learn developers (2026) *scikit-learn*, version 1.9.0 [computer software].
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- The pandas development team (2026) *pandas*, version 3.0.5 [computer software].
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